



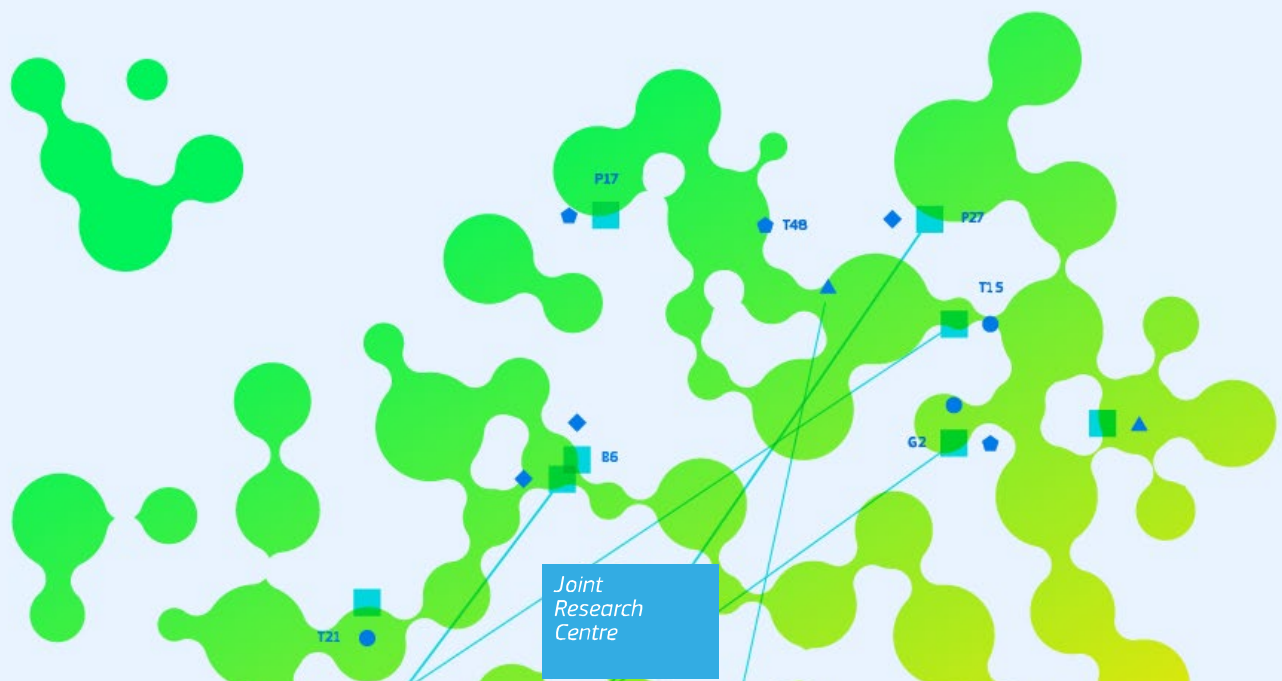
SMART SPECIALISATION STRATEGIES AND MISSION-ORIENTED APPROACH

Bridging theory and practice

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Abstract

The paper explores the extent to which established policy frameworks, particularly Smart Specialisation Strategies (S3s), can incorporate a mission-oriented approach (MOA). We build an analytical framework based on MOA key features and apply it to three case studies of S3s with a transformative ambition (Catalonia, Northern Netherlands, Brussels). Our findings reveal that shifting regional innovation strategies towards a mission-led or challenge-oriented dimensions entails trade-offs between strategic directionality and the bottom-up flexibility and face challenges in engaging societal actors in the governance, as well as in aligning policies and diversifying policy instruments beyond cohesion policy ones.

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Executive summary

This paper investigates the integration of a mission-oriented approach (MOA) within the Smart Specialisation Strategies (S3) framework. The broader goal is to explore how transformative innovation approaches aimed at tackling societal challenges can be effectively translated into regional innovation strategies. The study examines three case regions - Catalonia, Northern Netherlands, and Brussels - each of which has sought to incorporate transformative innovation principles into their respective S3s.

A mission-oriented approach shifts policy focus from strictly economic and technological objectives to addressing broader societal challenges through strategic, coordinated efforts. The study develops an analytical framework based on MOA principles and applies it to the selected S3s to assess their alignment with mission-oriented policymaking. The framework spans key dimensions of the MOA, namely, policy directionality, governance mechanisms, and policy instruments, to determine how effectively S3 can support mission-driven innovation.

Policy context

The paper fits into the ongoing discussion on the third framing of innovation policy, which focuses on addressing complex societal challenges rather than solely fostering technological advancements or market-driven innovation. It examines the barriers to implementing mission-oriented strategies within the existing Smart Specialisation framework and provides insights into the governance and policy adjustments needed to achieve systemic transformation. By evaluating regional case studies, the study contributes to the broader debate on how innovation policies can be structured to support long-term economic and societal objectives effectively.

Main findings

Findings reveal that while S3 provides a strong foundation for regional transformative innovation, its embedding of a mission-oriented or a broader transformative innovation approach faces various challenges. S3s increasingly exhibit a certain degree of directionality and intentionality towards societal challenges (SCs). However, the extent to which SCs are translated into specific, actionable priorities within S3s varies due to the difficulty to balance flexibility with directionality. This tension reflects the broader challenge of maintaining strategic focus without constraining innovation through excessive top-down control. Policymakers face the task of balancing directionality with the need for bottom-up experimentation. In response to this challenge, we find through the analysis of the case studies that S3s either adopt broader SC definitions or maintain separation between SCs and innovation areas to preserve flexibility in implementation.

Regarding governance, the study shows that, despite the commitment to broaden stakeholder involvement, it remains often unclear how the mechanisms for integrating societal actors into decision-making will work. While engagement with citizens and civil society is acknowledged, it is only to a limited extent embedded in structured governance processes. Policymakers struggle to move from theoretical commitments to practical inclusion, reflecting a broader challenge within mission-oriented frameworks. Despite ambitions for participatory governance, stakeholder involvement often remains aspirational rather than systematic. The effectiveness of governance structures is further influenced by regional institutional quality, administrative cultures, and relationships with civil society, which vary across contexts and can significantly impact implementation.

Finally, the study finds that existing policy instruments contemplated by the S3s do not yet fully align with the requirements of a transformative, challenge-oriented approach. While there is rhetorical commitment to synergising various funding sources, in practice, the dominant reliance on conventional funding mechanisms—such as the European Regional Development Fund (ERDF)—limits policy flexibility and innovation capacity. Many existing funding structures are not designed to support long-term, cross-sectoral, and experimental policy interventions, which are critical to MOA. Additionally, the fragmentation of funding and policy responsibilities across government departments further weakens the capacity for integrated and coordinated implementation. A key insight from the study is the disconnection between strategic transformative aspirations and the practical tools designed to achieve them. Cohesion policy funds, with their rigid structures and emphasis on cost-efficiency, may not provide the flexibility required for challenge-driven innovation.

1 Introduction

The emerging paradigm of ‘transformative’, ‘challenge-oriented’ or ‘mission-oriented’ innovation policies aimed at grand Societal Challenges (SCs) is subject to a growing body of studies (see Haddad et al., 2022). Yet the academic discussion within this strand has largely taken a theoretical perspective. Less attention has been devoted to how this new framing can be translated into concrete policymaking (Bergek et al., 2023; Magro and Wilson, 2019). The mainstreaming of challenge-oriented, mission-oriented or transformative concepts into policy practice demands more research. Otherwise, there is a risk of contradictory or fuzzy implementation, or even a drift towards ‘mission washing’ (Björk et al., 2022). The Horizon Europe missions provide an illustration of the sorts of challenges resulting from the current chasm between theory and practice. For instance, criticism about the lack of local legitimacy associated with the EU missions’ initiative (Gronchi, 2024; Cappellano et al., 2024) points to an issue that has been only partially addressed in the literature.

In order to address this research gap, one question that requires scrutiny is the extent to which established policy frameworks can accommodate a mission-oriented dimension. This paper strives to tackle this question by focusing on the specific case of Smart Specialisation Strategies (S3s). Smart specialisation is currently the most widespread approach regarding regional innovation in Europe, despite having been implemented to different fortunes (see e.g. Capello and Kroll, 2016; Gianelle et al., 2012; Kroll, 2015). It is therefore important to investigate how S3s can effectively bear a transformative change by integrating a Mission-Oriented Approach (MOA). This has been discussed only in a few papers and from a conceptual perspective. Scholars have pointed out that, despite having an implicit transformative orientation and directional traits, S3s have, to date, remained dominated by a narrow competitiveness-oriented and growth-enhancing interpretation of Research and Innovation (R&I) (Hassink and Gong, 2019; Tödtling et al., 2022). However, recent works on Sustainable Smart Specialisation Strategies (S4) and Partnerships for Regional Innovation (PRI) (Pontikakis et al., 2022) present the conceptual case for rethinking the S3 concept to embed a sustainable and inclusive dimension. The paper aims to complement this theoretical work by: a) defining the main criteria that should guide the incorporation of mission-led or transformative elements into regional innovation strategies; b) identifying potential bottlenecks and challenges in this respect by exploring three case studies of S3s with a transformative orientation.

Transformative, challenge-oriented and mission-oriented approaches are in many instances employed interchangeably, insofar as they refer to innovation policies targeting SCs. Nevertheless, they do not overlap, although one cannot find in literature a systematic discussion of their differences. Missions target specific, well-defined and time-bound goals which, in their mostly current understanding, may be conducive to systemic change. However, in its traditional version, which is still sometimes used, missions had a narrow technological and incremental focus (Robinson and Mazzucato, 2019). Transformative policies have the direct and inherent objective of generating systemic change, fundamentally restructuring social, economic, or technological systems to support long-term sustainability and equity. Challenge-led policies address broad societal issues, like climate change or public health, are also often used with the idea of changing the underlying systems (see for instance Tripl et al. 2024). Some authors, therefore, consider transformative innovation policies as an umbrella for mission oriented and challenge-driven policies targeting SCs (Haddad et al., 2022). The focus of the paper is the MOA, but its discussion applies by extension to

the other two approaches as the three are often conflated and overlapped both in the scholarship debate and policy practice.

The paper is divided into two sections. The first part begins with an epistemic conceptualisation of the MOA by defining some of its key aspects based on extant literature. The novelty of this analytical framework is that it is defined and compared to the specificities of the S3 approach casting light on how to reconfigure it in response to SCs. Thus, it takes into account the features and limitations of the smart specialisation concept, also from a policy practice perspective. The second part applies the framework to the evaluation of three case studies of mission-oriented/transformational S3s (Catalonia, Northern Netherlands and Brussels). In so doing, the paper strives to shed light on concrete obstacles and challenges resulting from operationalising the MOA into regional policy frameworks. The paper concludes with a policy recommendations on future S3s.

2 Smart specialisation and missions: commonalities and differences

The analytical framework provided in the following section rests upon three dimensions: policy directionality, governance and instruments. These three areas are grounded in policy practice but have also received emphasis in scholarly debates (see e.g., Borràs and Edquist, 2013; Cappellano et al., 2022). They match the dimensions considered by Larrue's (2021) categorisation of Mission-oriented Innovation Policies into strategic orientation, policy-coordination and policy implementation. In this section, the main features of the S3 and MOA in the three aforementioned areas are critically discussed against the previous literature on the topic. Originally, more dimensions, such as monitoring or legitimacy, were detected in the literature as essential elements of the MOA rhetoric. However, they were not taken into consideration in this study because they require a distinct procedural approach.

The aim of this section is to define the main differences and commonalities between MOA and S3 vis-à-vis the three areas. Our analysis does not focus on the smart specialisation concept, but on its operationalisation into the S3, that is, a defined policy framework which has been largely applied into regional contexts as part of place-based regional policies. On the other hand, we consider the MOA a more general theoretical approach that has been applied very differently at different scales and governance levels as well as in different places. Indeed, some scholars consider the smart specialisation strategies to be a specific application of the MOA, at least in its traditional technological meaning (e.g. Foray, 2018). However, it has been also noted that the smart specialisation approach is originally based upon a conventional idea of innovation policies focusing on economic and technological goals and does not contemplate a broader socio-ecological dimension in line with the new understanding of MOA (Hassink and Gong, 2019). Hence, we treat here the two concepts as separate although the comparison takes into account that one can be a subset of the other.

2.1 Policy directionality

The S3 introduced, or at least reinforced, the emphasis on the need to prioritise certain policy domains over others, through the Entrepreneurial Discovery Process (EDP): a policy practice to collectively assess the economic areas where the region has its own competitive advantage (Foray et al., 2009). The prioritisation and the result-oriented approach introduced by the S3 hints at what is called policy directionality. This has been defined as “promoting innovations that contribute to a particular direction of transformative change” (Parks, 2022). As a result, the notion of directionality has gained increasing salience (Weber and Rohrer, 2012) informing the whole discourse on modern innovation policies and nurturing a multitude of new policy approaches (see Haddad et al., 2022 for a deeper description of the concept of directionality). Among those, the MOA (Mazzucato, 2018) spearheaded the policy landscape with the launch of several mission-oriented policies at national and international levels (see the EU Missions).

While the S3 and MOA share some similar features in terms of policy approach, including the prioritisation, the stakeholder engagement, and the cross-sectorial ventures through better policy coordination (Foray, 2018; Mazzucato, 2018), they differ in their focus: S3 focuses primarily on economic competitiveness, while MOA is concerned with achieving significant outcomes with broad societal value (Reid et al., 2023). Foray (2018) argues that EDP can be part of a larger process

oriented at transformation as it should enable stakeholders to identify priorities which should be turned into a transformational roadmap. However, other scholars have downplayed the S3 potential to generate a transformative change, moving evidence from practical experience (Benner, 2022) as well as discussing its theoretical assumptions. Hassink and Gong (2019) raise criticism concerning the weak embeddedness of S3 in the regional innovation policy landscapes, the subdued inclusiveness of EDP which is mainly conducted by techno-economic stakeholders, adopting a very limited interest in different forms of innovation. The focus – hence, the directionality – is then limited to the economic competitiveness of the region. Tödtling et al. (2022) acknowledge that a new policy approach needs to be pursued to tackle SCs by including a larger and diverse set of actors, focusing on bright and dark sides of innovation, considering a whole set of innovation domains beyond the short-sighted perspective on Science, Technology and Innovation (STI).

The place-based focus of S3 should be also the result of locally run EDP, avoiding to “copy-paste” strategies produced elsewhere under a “one-size-fits-all” approach (Foray et al., 2009). Scholars have argued that a place-based contextualisation of SCs in both their framing and solution selection processes is essential to reap benefits of local market opportunities and gain public legitimacy on mission-oriented policy activities (Wanzenböck et al., 2020; Flanagan et al., 2023). The surge of what is called micro missions (Henderson et al., 2024) or local wins reflect an urban or local dimension of MOA based on very place-specific conditions.

2.2 Governance

The concept of smart specialisation assumes a shift in the governance of R&I away from the top-down models of the past (McCann and Ortega-Argilés, 2013). Influenced by systems of innovation theories, the S3 approach, as emphasised by Foray (2016), sees participatory and bottom-up processes as a critical factor to chart the right path towards a more competitive specialisation. To overcome the information limitations that have typically hampered traditional industrial policies, the identification of S3 priorities needs to draw upon and harvest (entrepreneurial) knowledge scattered across various actors (Hausmann and Rodrik, 2003). This requires, in the first place, a structured collaboration between policymakers, firms, research institutions and the civil society, through the EDP. At the same time, there is consensus in the literature on MOA around the involvement of a wide spectrum of stakeholders in the design and implementation of missions (Mazzucato 2018; Wanzenböck et al. 2020; Janssen et al. 2021; Larrue 2021; Wiarda et al., 2023). Given the contested and complex nature of SCs and their solutions, missions demand a great deal of coordination and legitimacy across many actors (Lindner et al., 2021).

Such a complex, multi-agent governance is not exempt from theoretical and practical limitations. A certain degree of ambiguity relative to the scope of S3's EDP, a problem first raised by Capello and Kroll (2016), has resulted in diverse, often relatively narrow interpretations and applications of the concept under the influence of pre-existing institutional configurations, governance arrangements, and economic circumstances (McCann and Ortega-Argilés, 2013). Thus, in many instances, the EDP has fallen short of achieving a satisfactory involvement of the quadruple helix actors (university-industry-government-civil society), in particular from the civil society actors, as highlighted in various studies (e.g. Aranguren et al., 2018), which contrasts with the need for broad public engagement underpinning the MOA approach. Moreover, within both the S3 and the MOA, the capacity and reputation of institutions is seen as critical to building shared visions (Benner, 2019). In the context of MOA, there is also an emerging discussion about the inherent difficulty in

reconciling directionality and subsidiarity in multi-governance, multi-party settings (Uyarra et al., 2023; Grillitsch et al., 2023).

Experimentalism is, at least on paper, a key dimension in both MOA (Mazzucato, 2018) and S3 (Radošević et al., 2017). Because of the inherent uncertainty of innovation processes, it is appropriate to adopt a flexible and recursive policymaking constantly adjusting to the information emerging from broad interactions and the testing of different approaches. However, various works have highlighted that the scope for experimentalism in S3 has been, to date, severely restricted by the rigid, audit-driven, and ultimately cost-efficient, nature of cohesion policy rules (Gianelle et al., 2020; Rauhut and Humer, 2020), as well as by the disproportionate emphasis on design, lack of policy capacity (Laranja, 2022), and institutional arrangements (Morisson and Doussineau, 2019). The latter two problems are a common issue in MOA. Additionally, the need to set clear objectives in the MOA can clash with the “muddling-through” approach – the idea that policies evolve through trial and error, learning from small-scale changes rather than making grand, long-term plans – which underlies experimental policymaking; at the same time, the emphasis on experimentalism under both the MOA and S3 raises issues in terms of institutional accountability (Radošević et al., 2023).

2.3 Policy instruments

The selection of policy instruments is also paramount in innovation policies. Foray (2023) argues that a bad engineering of instruments can cancel out even the most well-thought innovation policies. This is even truer in relation to the design of policy mixes, which, in the context of S3, can turn out to be a very complex exercise implying trade-offs and tensions considering its operationalisation in multi-level settings and pre-existing governance, policy and institutional arrangements (Reeid and Maroulis, 2018; Nauwelaers et al., 2014). Surprisingly, it seems that this dimension has attracted less attention from scholars than the other dimensions of S3. However, anecdotal evidence seems to show that the use of policy mixes has been largely limited for S3s. In fact, in spite of the initial ambition to act as overarching policy frameworks, many strategies have ended up guiding only the R&I investment of European Structural and Investment Funds (ESIF) (Pontikakis et al., 2022). Even in this case, Gianelle et al. (2020) found that in a non-negligible number of cases there is little alignment between the S3s and policy interventions under ESIF whilst D’Adda et al. (2021) found scant evidence that S3s have re-shaped ESIF funding decisions.

The complex and cross-sectoral nature of SCs addressed in the MOA requires a strong integration across various instruments (Larrue, 2021). This idea dates back to first works referring to mission-oriented policies. Mazzucato (2021) stresses the importance of using a wide array of financial instruments, including more innovative ones such as innovative public procurement (e.g. Merisalo et al., 2024) and challenge prizes (e.g. Morrison et al., 2023), to cater for different funding needs and risk profiles. Moreover, given the systemic perspective of missions, supply-push instruments need to be integrated along with demand-pull policies (Tödtling et al., 2021).

3 Description of the analytical framework

Using the three dimensions discussed in the previous literature review, this section develops an analytical framework (Table 1) for assessing S3s vis-à-vis MOA and SCs. The evaluation framework defines therefore the optimal characteristics that a mission-led S3 should have.

Firstly, we explore the notion of policy directionality which, as described in the previous section, is a quintessential tenet of the new wave of innovation policy approaches including MOA. However, policy directionality might remain a claim if not substantiated with clear answer to: what; why; how; and when? With this line of reasoning, the proposed analytical framework unpacks the concept into criteria which underpin evaluative questions. Resonating with MOA's assumption to divert the focus of innovation policy from the rate of growth to the route (Mazzucato, 2018), it is critical to understand which directions are promoted through S3s and whether they are aligned with SCs. Such directionality should not be considered fixed in space nor in time. In fact, it is reported to depend on the way policymakers want to navigate the problem-solution space (Wanzenböck and Frenken, 2020), which is time-dependent and can be framed across goal, sectoral, and solution dimensions (Andersson and Hellsmark, 2024). Policymakers need to ensure alignment across policy objectives, logic, domain, and leverage (Bergek et al., 2023) navigating trade-offs and conflictual interests across the actors involved. The practical experience arising from evaluative studies shows that many policymakers take EDP as a "tick-box exercise" rather than adjusting their mindset and operational practices to a more entrepreneurial approach (Benner, 2019).

The policy directionality is assessed based on the degree of focus on specific SCs in the S3s: more specifically, based on the extent of granularity in the analysis and objectives of each SC provided by the S3. SCs should be considered as multi-nature and multi-dimensional (Mazzucato et al., 2020). Their implications open different scenarios with inherent trade-offs across economic, environmental, and social effects that need to be navigated by policymakers. When assessing policy directionality, the breath of analysis considering this aspect will work as a proxy for the degree of focus towards SC in an S3. To evaluate the policy directionality, it is also important to focus on the evidence-based support reported in the S3 to justify the direction chosen. An S3 usually builds on a regional diagnosis (Foray, 2016). If the S3 is driven toward SCs, this diagnosis should carry also empirics concerning the effects or the threats generated by the SC. The framework equally considers the alignment of objectives and funding instruments as a critical lens to evaluate policy directionality. If proved to be mission-oriented, the analysis would yield evidence on the coherence of objectives, actions, and expected outcomes across multiple sectors, pursued with/by a set of different actors aligned toward a commonly defined mission-oriented target.

An authentic S3 policy should pursue mission-oriented objectives that are realistic and tailored to place-specific conditions, considering local risks, demographics, cultural values, social fabric, and economic and workforce capacities. The policy's directionality must adapt over time to changing circumstances, ensuring consistency through well-defined objectives and actions. Evaluating the granularity of interventions is crucial to assess the mission-orientation of the regional S3. The second dimension of the framework is the governance. This requires looking into whether the governance arrangements depicted in the S3s are designed to prevent coordination failures. The risk of such failures in the application of the MOA is not insignificant as the cross-sectoral nature of S3s may entail the involvement of multiple departments or agencies, and diverse non-state agents (Uyarra et al., 2023). In principle, a mission-oriented S3 should provide a description of the governance framework and mechanisms through which all these actors collaborate and contribute

to the related policymaking processes (e.g. identifying opportunities, defining priorities, monitoring, etc.).

This means going beyond a simple explanation of the consultation process in the design of S3s, as it has been largely the case with the application of the EDP (Radosevic et al., 2023), and articulating how the involved parts are expected to participate in the implementation in a structured fashion. An additional element to assess is, whether the S3 details what specific configurations of actors are associated to each of the identified SCs and how they are expected to interact within these configurations. Here the reference is the research on mission-oriented innovation systems (Hekkert et al., 2020) or less complex settings such as the CHallenge-Oriented Innovation paRtnerships (CHOIRs) (Pontikakis et al., 2022). The most pivotal dimension in relation to the governance of S3 is the breadth of involvement, particularly in relation to the civil society at large. Specifically, a mission-oriented S3 shall be assessed against the capacity to define governance arrangements that, alongside the usual actors of the innovation system, encompass on an equal footing citizens, civil society organisations, professionals (Wanzenböck et al., 2020).

Analysing the experimentalist dimension is a less straightforward exercise. One important aspect is that the governance arrangements defined in the S3 should work in a way that allows for a regular revision of its goals and implementation trajectories (Lepore and Spigarelli, 2018) based on bottom-up feedback and (collective) learning. This requires assessing if the S3 governance envisages viable mechanisms for continuous collaboration, learning, embedding of new information into the process and ultimately adjustment of policymaking. Crucially, the different stages of the policy cycle, that is, consultation, design, implementation and monitoring and evaluation, should come in an integrated and simultaneous process, so as to identify and remedy to errors in due course, rather than being conducted separately and sequentially.

The third and final dimension to assess embedding of the MOA in S3s concerns the policy instruments (Borràs and Edquist, 2013). Two aspects need to be probed in regard to which instruments are identified in the S3 to advance its goals. The first one is whether the S3 presents a holistic approach in this area, giving at least a sense of how the proposed mix works in a coordinated way. This does imply that the S3 cannot content itself with referring to the European Regional Development Fund (ERDF) only, as it has been often the case. The second criterion lies with the variety of instruments selected by the S3, not just in terms of different financial tools, but also non-financial ones (e.g. regulatory sandboxes). As explained above, using a diverse set of instruments is important to tap into different funding needs, some of which cannot be addressed by traditional funding sources (for instance the ERDF, given its project-driven and cost-efficient orientation) (Mazzucato, 2016).

Table 1. Analytical framework

Dimension	Criterion	Question(s)	References
Policy directionality	The extent of S3 focus on SCs	Does the S3 mention specific SCs as key areas of development?	Foray (2018); Parks (2022); Larrue (2021); Wanzenböck et al. (2020); Elzinga et al. (2023); Bergek et al. (2023); Hassink et al. (2022)
	The alignment of objectives and funding instruments	How do the SCs/missions fit in the strategic orientation of the S3 (especially vis-à-vis innovation/research areas)?	
	Sense-making of SC	How much are the SCs contextualised?	Wanzenböck and Frenken (2020); Uyarra et al. (2023)
	Evidence-based support of challenge-oriented directionality	Does the S3 explain how the capacities of the region help to tackle the SC(s)?	
	Consistency across analysis, objectives, actions	Do the proposed actions (explicitly) state how they help to solve the SC(s)?	Kirchherr et al. (2023); Andersson and Hellsmark (2024)
Governance	Extent of participation and involvement of societal actors outside the innovation system	Does the S3 describe (concretely) a wide participation of the society for solving the SC(s)?	Foray (2018); Parks (2022); Larrue (2021); Wanzenböck et al. (2020); Elzinga et al. (2023); Bergek et al. (2023); Hassink et al. (2022)
	Governance mechanisms	Does the S3 provide a comprehensive description of governance framework based on viable arrangements for multi-stakeholder coordination and participation throughout the whole policy process? Does the S3 spell out what specific configurations of actors are associated to each of the identified SCs and how they are expected to interact within these configurations?	Wanzenböck and Frenken (2020); Uyarra et al. (2023); Wanzenböck et al. (2020)
	Experimentation	Does the governance of the S3 allow for real-time revision and adjustment of the policy?	Radosevic et al. (2023)
Policy instruments	Breadth of policy mix	Does the S3 describe how the policy instruments work in a coordinated and integrated manner?	Larrue (2021)
	Variety of policy mixes Experimentation	Does the S3 present a diverse mix to address different funding needs, including non-financial tools?	Mazzucato (2021); Borràs and Edquist (2013)

Source: Authors' elaboration

4 Case studies

4.1 Methodological approach

In this section the analytical framework defined above is applied to the three regional S3s: Catalonia, Northern Netherlands, and Brussels. The primary criterion used for the selection of the three case studies has been their explicit anchoring in a mission-oriented, challenge-oriented or transformative innovation paradigm. To the best of our knowledge, a certain number of other S3s reflect in various ways approaches hailing from these paradigms. However, the pool of S3s offering a mature and fully-fledged operationalisation of such concepts, thus making them suitable to this analysis, is far more limited². This limitation has affected the geographical representativeness of our sample. The three case studies are from regions with substantial differences. Catalonia and Brussels are metropolitan, high-income regions with strong innovation capacities, while the Northern Netherlands, comprising Drenthe, Friesland, and Groningen, is rural, with a below-average Gross Domestic Product (GDP) and a fragmented innovation sector despite improvements. Catalonia and Brussels are knowledge-intensive hubs, whereas the Northern Netherlands has faced challenges from industrial decline and energy transition due to its traditional focus on manufacturing, gas extraction, and agriculture. Governance structures also vary: Belgium's federal system grants Brussels budgetary autonomy and exclusive competences on economic development where the latter are shared under Spain's asymmetric decentralisation. Dutch provinces have limited powers, mostly executive. Further, the governance of cohesion policy funds, important for S3s, differ in the three areas: competences for funding programmes are shared between region and state in the case of Spain, shared among different provinces in the case of the Netherlands, and exclusively regional in the case of Belgium.

The analysis was conducted through a review of the respective S3s for the period 2021–2027, including annexes. We also conducted semi-structured interviews with both policy-makers and experts from the European Commission, EU funding programmes and research institutes. The interviewees have been selected on the basis of their significant experience and knowledge of both S3s and MOA, and because they have been involved extensively with various roles in the design, supervisions or evaluation of S3s. The methodological choice presents drawbacks because it does not allow to collect more in-depth information on some aspects that might be detailed in more operational documents. The framework has been nonetheless designed taking into account these limitations inherent to the format of a strategic document. From a methodological perspective, in line with other qualitative studies conducted through document analysis (Bowen, 2009), the rationale behind evaluation of the data was shared among all the research team. The evaluation of the documents was developed in a two-step's procedure: first, authors individually assessed the documents against the criteria in Table 1. Thereafter, the authors jointly discussed their findings to

² We did screen other S3s to widen our geographical scope and particularly considered the S3s of Pirkanmaa, Tuscany and North-East Region (Romania) but after a careful consideration deemed them less fully-fledged than the included three cases and, thus, ill-suited for our purposes. Similarly, we excluded the S3 of the Czech Republic outside the scope of our analysis, since, while it has a clear mission-orientation, it is a national, not a regional strategy.

increase the assessment's coherency across the research team. This method allowed to have a wider perspective on the documents assessed. That is, we followed the standard procedures of qualitative content analysis (Vears and Gillam 2022). First, we reviewed the policy documents to gain an interpretative understanding of their content. Using our analytical framework, we identified and coded text passages discussing policy directionality, governance, and instruments. We compared notes, refined categorisations through second-round coding, and finalised the textual data. Simultaneously, we assessed and made minor adjustments to the framework. Finally, we analysed the data, synthesised key messages, shared findings, and reached a consensus.

The interviews had a three-fold purpose: first, to discuss and validate the conceptual robustness of the analytical framework; second, to discuss the results of our analysis, further contextualising and triangulating them with the expertise of the respondents; third to test the results with experts who have direct knowledge of implementation of S3s. A summary of the comparative findings is presented in annex 1.

4.2 Policy directionality

Since, the three S3s were picked for their declared transformative ambition, it is no surprise that they all present a relatively clear identification of relevant SCs. In the case of Catalonia SCs are even explicitly understood and articulated as (transformation of) “socio-technical systems” (e.g. a sustainable, fair, equitable and healthy food system). As far as the Northern Netherlands is concerned the SCs are called “transition tasks” and linked to relevant Sustainable Development Goals (SDGs) as well as Europe’s missions (under Horizon Europe). This alignment not only clarifies the strategic orientation of the S3 but also strengthens its transformative ambition by directly connecting local challenges to overarching, internationally recognised goals. In the three strategies, SCs are the core part upon which the documents are built, thereby informing their overall strategic orientation in all three cases. This marks a fundamental shift from the traditional S3 approach which is rather organised around innovation areas and technologies. That is still the case for several existing S3s, even for those where SCs, or references to them, are included. Conversely, innovation domains are framed in the three S3s as instrumental to the SCs instead of being treated as goals per se. The S3 from Brussels, in particular, offers an in-depth explanation as to how the innovation domains contribute to each SCs. In the case of Catalonia the innovation areas and SCs do coincide but the specific objectives of the SCs in this S3, remain vague. This appears to be a deliberate strategic choice to leave more flexibility in the gradual shaping of the SCs and their objectives, by means of so-called ‘shared agendas’, throughout the (participatory) process. Nonetheless, it somewhat weakens the directionality within the S3. The “sense-making” of the SCs, that is, an explanation of why they are particularly relevant for the regional context and contribute to shaping it, is provided in the three S3s, including data and projections/forecasts. In addition, Northern Netherlands spells out the development opportunities for the region arising from addressing the SCs. Finally, only the S3 from Brussels offers a more detailed description of actions to address SCs.

By providing concrete action plans for each SC, the strategy translates strategic intentions into tangible goals and steps. This focus on clarity and structure within Brussels’ approach significantly strengthens its directionality by directly linking resources, actors, and outcomes to specified societal needs. The (partial) lack of this element in the other two S3s might be due to the intrinsic format of the document or the reluctance to commit to solutions in advance given the uncertainty of the

innovation process and leave room for experimenting different paths. In all three cases, SCs are mostly approached from the supply-side angle. Despite demand articulation—such as creating public demand for sustainable solutions—can be a powerful lever for transformation, it is only vaguely referenced, if at all, in the documents. The supply-focused approach, common in many traditional innovation policies, may limit the full potential of SC-oriented strategies to address societal needs comprehensively, as demand-side policies are crucial to fostering adoption and scaling innovations aimed at societal benefits.

4.3 Governance

All three S3s extend governance to the entire quadruple helix, including societal actors often considered to lay outside the innovation community, crucial for mission-oriented policies. The role of civil society is more detailed in Northern Netherlands but addressed broadly in Catalonia and Brussels, where both emphasise involving groups most affected by the SCs. However, the integration of societal actors into governance processes is not entirely clear. For instance, Catalonia's Opportunity Discovering System relies on working groups formed by quadruple helix actors, though details on their composition are lacking. Governance descriptions are more accurate in Catalonia, which adopts a flexible shared-agenda approach for experimentation. Northern Netherlands favours a bottom-up model, including knowledge and entrepreneurial representatives in its steering committee, contrasting with Catalonia's government-led governing body. Yet, specifics on actors involved in each SC remain vague across all cases.

Similarly, Northern Netherlands emphasises the need for flexible governance arrangements that encourage experimentation, especially through its strong focus on stakeholder engagement. This adaptive stance allows the region to respond to new challenges dynamically, though the limited involvement of civil society in formal governance structures may impede truly comprehensive adaptation. Brussels' governance model incorporates mechanisms for real-time adjustment and feedback, aligning with the experimentalist ethos that characterises mission-oriented policies. The need for a reflexive approach leading to "adjustments" is acknowledged in all the three S3s. This element alone, alongside the focus on how bottom-up approaches apply to the implementation of the S3, not just the design, marks another fundamental difference with previous S3s.

4.4 Policy instruments

The approach to policy instruments in the three S3s is the least aligned "dimension" with expectations arising from the analytical framework. Two out of the three S3s (Catalonia and Northern Netherlands) analysed provide only an indicative and generic mention of the instruments they plan to employ but fall short of providing more information. Only the Brussels' S3 offers a more detailed account in a standalone section, including a short description for each instrument and potential beneficiaries. There are two main areas where the S3s do not articulate an approach to instruments as required by a fully-fledged transformative and challenge-oriented ambition. First, the proposed mix is essentially composed of financial mechanisms only, with some exceptions in

the Brussels' S3. Second, the S3s are largely referring to instruments under the remit of EU programmes (Catalonia and Northern Netherlands). This reliance on EU cohesion funds highlights a common constraint in regional innovation strategies, where these funds have very rigid rules and funding criteria, limiting the flexibility and experimentation necessary for addressing SCs effectively. In our interviews it was noted that this leads to apply a project-based logic to tackle SCs rather than more holistic strategies.

The S3 from Brussels contemplates only the budget of Innoviris, the regional innovation agency. There is little mention of other funding sources. It must be noted that in the S3s from Catalonia and Northern Netherlands there is emphasis on creating a synergistic approach across different EU funds. Thirdly, effective implementation of mission-oriented policies relies on integrating and coordinating a range of policy instruments to ensure they work in harmony toward SCs. However, the three S3s offer little information as to the levels of integration across their policy mixes. The Brussels strategy lacks details on coordinating and integrating its policy instruments effectively. While Northern Netherlands mentions cross-cutting themes to connect SCs, it remains vague on specific instruments, and Catalonia provides no clear guidance on which instruments will support its strategy's implementation. Finally, despite a prevalent supply-side focus, demand articulation instruments are also involved in the form of public procurement for innovation.

5 Discussion

Conceptually, the findings underlines that all the three S3s point – to different extents – to a definite directionality/intentionality towards specific SCs. SCs are claimed to be at the core of the documents, thereby informing their overall strategic orientation in all three cases. However, despite the overall match between the R&I areas and the SCs, the policy objectives, except for the Brussel's strategy, have a more nuanced definition. This reflects the existence of inherent trade-offs between flexibility and directionality in the challenge-oriented or mission-oriented approaches. The analysis of the S3s hint that policymakers are faced with the difficult task of balancing the strategic (top-down) focus required by directionality and the flexible (bottom-up) nature of its implementation. This problem has been explored, albeit from different angles (directionality versus policy coordination or subsidiarity), in some recent works (Grillitsch et al., 2023; Uyarra et al., 2023). It also foreshadows a more fundamental quandary in relation to exerting directionality without constraining too much both the curiosity-driven and market-oriented drivers of the innovation process (by pushing it towards expected politically driven outcomes). This might hurt both freedoms to experiment and competitiveness in contrast with the smart specialisation core objectives. Hence the decision in the S3s to solve the potential misalignment between SCs and innovation areas either by establishing broad SCs or by keeping them separate.

Concerning governance, despite the ambition in all three cases to ensure a wide range of involved stakeholders, the strategies are vague concerning how societal actors will specifically fit in the governance processes. Citizens are involved in single activities often linked to projects rather than through structured co-creation strategy process. As noted by one expert, citizens engagement is a challenge in many mission-led frameworks, including EU missions.

In addition, the capacity of different regional governments to implement participatory processes open to the civil society depends on a number of factors (e.g. quality of institutions, administrative cultures, institutional reputation, relations with civil society organisations) which are not directly related to the participatory mechanisms themselves. Thus, even a detailed description of engagement mechanisms does not warrant the achievement of a successful participation. Still, concerning governance and flexibility, the three S3s can be considered a step forward in the inception of MOA to implement bottom-up experimentation.

Finally, the three cases do not present an approach to policy instruments as required by a fully-fledged transformative and challenge-oriented rhetoric. Despite the pledge in the strategies to pursuing a synergistic approach across different (EU) funds, the information on funding and policy instruments does not allow to scrutinise a progress from the traditional S3 implementation, which in most instances narrowly focus on ERDF, toward a modern SC-oriented innovation approach. One related issue emerging from the interviews is that the design of ERDF measures themselves do not mirror a mission-led approach which further hinders its operationalisation. There is also scant evidence of an integrated approach spanning different policies which is critical to MOA. While the new generation of S3s refer more explicitly to other policy areas, they provide little details in regards to their alignment with STI policies, which certainly hampers the implementation of a MOA approach, as noted by the interviews. One expert highlighted that, even though mission-oriented or similar approaches in S3s are designed to have cross-sectoral approaches, their implementation will follow a siloed sectoral approach mainly based on projects. Linking S3 mostly to EU cohesion policy funds, because of their cost-efficient logic and complex provisions, may challenge the application of SC-oriented approaches by limiting experimentation, flexibility, and the adoption of diverse, long-

term, and failure-tolerant instruments. As put by one interviewee, 'there is a disconnection between what you would like to achieve and what your tools can allow you to do.' Additionally, the pressure for absorption on the side of cohesion policy funds might not sit well with the experimentation required by a challenge-oriented S3. There is also an issue linked to the fact that cohesion policy funds and S3s are managed by different departments within many regions.

6 Policy implications and conclusions

This paper explores the specificities of the S3 approach casting light on how to reconfigure it in response to SCs. The rationale for this approach lies at the intersection of theoretical arguments and practical evidence. The paper hinges on an analytical framework based on three main pillars of the MOA conceptualisation. Starting from a theoretically informed analytical framework, the authors assessed real-world cases of S3s implemented in three EU regions that claimed to tackle SCs (or Sustainable Development Goals - SDGs). The empirics, feed into the conceptual debate around MOA as well as providing practical takeaways for policymakers. Our results underscore the following policy implications.

First, there is a growing number of S3s embedding mission-led or transformative elements, amongst which the three analysed are arguably among the most advanced ones. Yet the S3 conditionality requirements and guidelines elaborated by the European Commission remain still based on a competitiveness framing of R&I policies. This suggests their revision in the future towards a mission-led direction, which would be critical to bring clarity in the operationalisation of these approaches against the non-negligible risk of mission-washing or fuzzy and diverging implementation of transformative innovation aspects in future S3s. The latter risk is appearing, albeit marginally, in some results of our analysis, whereas aspects that could be considered as mission washing have been encountered in some S3s screened when identifying our case studies. Second, our analysis suggests that S3s tend to maintain a narrow focus on cohesion policy funds. This limits the scope for experimentation and the use of diverse instruments, which in turn is pivotal to advance a MOA perspective. This limitation might be addressed at least partially by legally requiring S3s to articulate how they plan to use other funding sources in synergy with cohesion policy ones. Another requirement could imply stating what non-funding instruments will be used or leveraged. Third, there is a need to reinforce policy learning/transfer mechanisms, for instance peer-learning programmes or platforms for knowledge exchange, to help regional governments develop approaches to address the limitations identified in the three case studies.

Finally, this analysis presents some limitations. First, it focuses only on three more developed regions. Second, the analysis does not complement and corroborate information extracted from the textual analysis of S3s with other research approaches, for instance by surveying the staff of the regions involved in the design of the S3s. Hopefully, these limitations can be tackled in future research.

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List of abbreviations and definitions

Abbreviations	Definitions
CHOIRs	Challenge-Oriented Innovation paRtnerships
EDP	Entrepreneurial Discovery Process
ERDF	European Regional Development Fund
ESIF	European Structural and Investment Funds
EU	European Union
GDP	Gross Domestic Product (GDP)
MOA	Mission Oriented Approach
PRII	Partnerships for Regional Innovation
R&I	Research and Innovation
SC	Societal Challenges (SCs)
S3	Smart Specialisation Strategies
S4	Sustainable Smart Specialisation Strategies
SDGs	Sustainable Development Goals
STI	Science, Technology and Innovation

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Annexes

Annex 1. Summary of findings per region

Table 2. Summary of the findings for Brussels, Northern Netherlands and Catalonia

Dimension	Criterion	Brussels	Northern Netherlands	Catalonia
Policy directionality	The extent of S3 focus on SCs	Improved resilience and prosperity are set as key priorities that can be achieved by contributing to solving SCs	The strategy is built around specific SCs	SCs are mentioned as key development targets, but the strategy does not formulate exact SC-related objectives
	The alignment of objectives and funding instruments	SCs are associated with strategic areas of innovation to clarify their potential contribution to mitigate the SCs	There is a good fit between the SCs and the strategic orientation of the strategy but potential links between them and innovation areas remain fuzzy	The strategy detects opportunities for synergies and complementarities between European funds and programmes to support the initiatives for SCs
	Sense-making of SCs	The impacts of the SCs are considered upon the regional specificities	The strategy contextualises the transition goals and how they offer development opportunities for the region	The contextualisation of SCs is somewhat vague
	Evidence-based support of SC-oriented directionality	The strategy describes the STI, economic and sectoral development of the region in reasonable detail	The strategy describes why the region is in a favourable position to tackle the specific SCs	The strategy references various regional documents where to find this information
	Consistency across analysis, objectives, actions	The actions discussed are generally quite explicit in how they can potentially produce solutions to the SCs	Generally, the chain of development remains ambiguous but, in some instances, explicit solutions are mentioned	There is no meaningful information about the defined actions

Governance	Extent of participation and involvement of societal actors outside the innovation system	The drafting of the strategy included a broad consultative component including quadruple helix actors that are also considered as key actors to solve the SCs	The strategy stresses that the participation from all quadruple helix actors is needed for solving the challenges.	There are some instruments to collectively identify SCs and to consider their long-term effects
	Governance mechanisms	The actors associated with the SCs are listed but their exact role is not spelled out	The specific configurations of actors is missing (e.g., civil society organisations are not mentioned)	Overall, the description of the governance is detailed but there is a lack of clarity in the participation of stakeholders in the governance
	Experimentation	The governance model incorporates mechanisms for real-time adjustment and feedback	The strategy strives towards recognising development opportunities in a timely manner	The strategy promotes experimentation with possible solutions to address SCs involving the affected groups
Policy Instruments	Breadth of policy mix	The strategy does not describe how the policy instruments work in a coordinated and integrated manner	There are cross-cutting themes that link the SCs together but overall, the strategy remains vague about policy instruments	Overall, there is no clarity as to which policy instruments are expected to support the implementation the strategy
	Variety of policy mixes Experimentation	The strategy concentrates on describing the funding from Innoviris but not from other sources	Despite being a little vague, the strategy stresses the importance innovative funding instruments but there is no mention of non-financial instruments	There seems to be a narrow focus on EU funds and there is no direct reference to non-financial resources

Source: Authors' elaboration

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